

Zhao, Yuxuan

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EDUCATION

University of Illinois Urbana-Champaign Urbana IL USA PhD Candidate in Mathematics	Aug 2023 – Present
ShanghaiTech University Shanghai China BS in Mathematics and Applied Mathematics	Aug 2019 – Jul 2023

RESEARCH EXPERIENCE

Doctoral Research University of Illinois Urbana-Champaign Advisor: Prof. Daniel B. Cooney Research Interests: Mathematical Biology, PDEs, Dynamical Systems - Conduct research on nonlinear dynamical systems and PDE models using stability, bifurcation, and asymptotic analysis. - Develop numerical simulations and computational pipelines in Python/MATLAB for parameter sweeps, pattern formation, and model validation.	Dec 2024 – Present
Undergraduate Research ShanghaiTech University Advisor: Prof. Shixiao Jiang Thesis: <i>Some Meshless Methods on Smooth Manifolds</i> Implemented and reproduced meshless numerical methods for solving PDEs on smooth manifolds, including local RBF-FD and local MLS methods.	Dec 2022 – May 2023

PUBLICATIONS AND MANUSCRIPTS

- Finite-Wavelength Pattern Formation in a Nonlocal Self-Chemotaxis Model (Manuscript in preparation). - The Keller-Segel Model for Chemotaxis on Metapopulation Graphs and Graphons (Manuscript in preparation). Zhao, Y., Zhu, K., Zhang, Y., Cooney, D. B. Pattern Formation in a Spatial Public Goods Dilemma due to Diffusive or Directed Motion. <i>Bulletin of Mathematical Biology</i>, 88, 173 (2026). doi:10.1007/s11538-026-01733-0.	
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RESEARCH MENTORSHIP

IML Project Mentor (Volunteer) University of Illinois Urbana-Champaign Urbana, IL USA - Mentored undergraduate students interested in mathematical research through the Illinois Mathematics Lab (IML). - Designed and supervised small-scale research projects suitable for undergraduate researchers. - Worked under the guidance of faculty instructors, serving as the graduate mentor for student research teams.	Jan 2026 – May 2026
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TRAINING PROGRAM

INMAS Training Program University of Illinois Urbana-Champaign Urbana, IL USA - Selected for competitive industrial math & statistics training program. - Applied Python, statistics, and machine learning to interdisciplinary data-driven projects. - Delivered technical presentations to peers and mentors.	Oct 2024 – May 2025
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WORK EXPERIENCE

Teaching Assistant and Grader University of Illinois Urbana-Champaign Urbana, IL USA Calculus, differential equations, machine learning, real analysis, and topology.	Aug 2023 – Present
PrairieLearn Summer Intern PrairieLearn, Inc. Urbana, IL USA - Developed Python-based interactive assessment and auto-grading systems using NumPy and SymPy. - Collaborated with a cross-functional team on implementation and deployment.	May 2025 – Aug 2025

SKILLS & INTERESTS

Languages: Fluent in English and Mandarin Chinese	
Technical Skills: Python (NumPy, Pandas, SciPy, SymPy); MATLAB; $\text{\LaTeX}$; Git; Microsoft Suite	
Interests: Hiking, table tennis	